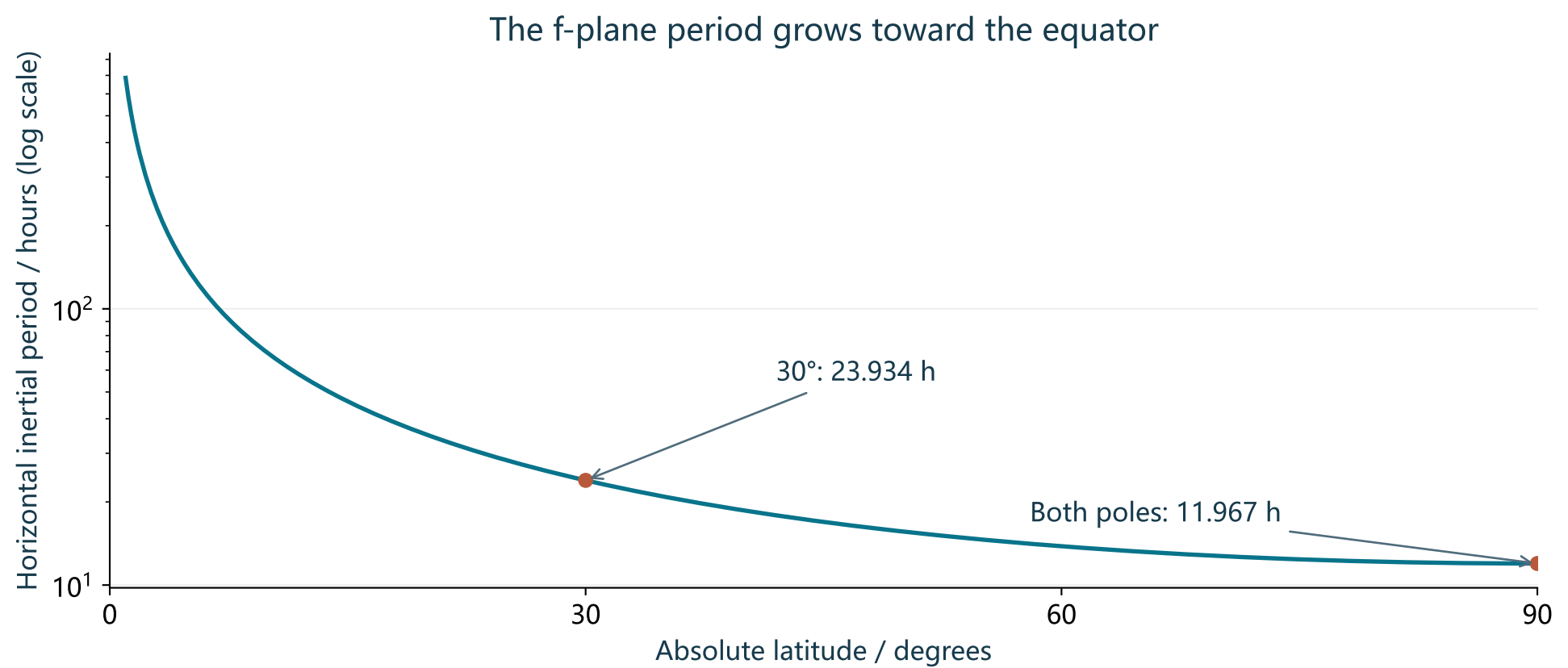

At f=0 there is no finite-period horizontal inertial circle in this model; other equatorial motions remain possible.